

SAT UZBEKISTAN

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Module-1

1



Mark for Review



In triangle ABC , the measure of angle A is 34° and the measure of angle B is 90° . What is the measure, in degrees, of angle C ?

(A) 34

(B) 56

(C) 90

(D) 180



Which expression is equivalent to $3xy(2x^2 + 7y)$?

(A) $5x^3y + 10xy^2$

(B) $6x^2y + 7y$

(C) $6x^2y + 21xy$

(D) $6x^3y + 21xy^2$



Which expression is equivalent to $30x^2 + 6x + 6$?

(A) $5x^2(6x + 6)$

(B) $30x(x + 1)$

(C) $6(5x^2 + x + 1)$

(D) $4(2x^2 + x + 1)$



For a particular tree species, it is recommended to plant a maximum of 92 trees per acre. Based on this recommendation, what is the maximum number of trees of this species that can be planted on 2 acres?

(A) 276

(B) 184

(C) 94

(D) 90

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ $3\frac{1}{2}$
$\frac{2}{3}$	$\frac{2}{3}$.6666 .6667	0.66 .66

5



Mark for Review

A gym charges its members a onetime \$34 enrollment fee and a membership fee of \$16 per month. If there are no charges other than the enrollment fee and the membership fee, after how many months will a member have been charged a total of \$178 at the gym?

Answer Preview:



$$f(x) = 7x + b$$

For the linear function f , b is a constant and $f(8) = 56$. What is the value of b ?

(A) 0

(B) 1

(C) 7

(D) 8



The function f defined by $f(t) = 14t + 9$ gives the estimated length, in inches, of a pothos plant t months after Shruti purchased it. Which of the following is the best interpretation of 9 in this context?

- (A) Shruti will keep the pothos plant for 9 months.
- (B) The pothos plant is expected to grow 9 inches each month.
- (C) The pothos plant is expected to grow to a maximum length of 9 inches.
- (D) The estimated length of the pothos plant was 9 inches when Shruti purchased it.



At a convention center, there are a total of 375 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.48, and the probability of selecting a visitor who is located in room B is 0.24. How many visitors are located in room C?

(A) 28

(B) 43

(C) 90

(D) 105



$$x^2 - 28x = 0$$

Which of the following is a solution to the given equation?

(A) 56

(B) 28

(C) 14

(D) $\sqrt{28}$

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666	0.66 .66 0.67 .67



10



Mark for Review

A right square pyramid has a height of 12 inches and a volume of 196 cubic inches. What is the perimeter, in inches, of the base of this pyramid?



The value of a share of stock was \$240 on the day it was purchased. Every year, the value of the share of stock increases by 1% of its value the previous year. Which function V gives the value, in dollars, of the share of stock t years after the day it was purchased?

(A) $V(t) = 240(1.01t)$

(B) $V(t) = 240(1.1t)$

(C) $V(t) = 240(1.01)^t$

(D) $V(t) = 240(1.1)^t$



If $f(x) = x + 3$ and $g(x) = 3x$, what is the value of $6f(3) - g(3)$?

(A) -3

(B) 12

(C) 18

(D) 27



$$3x + 9y = 10$$

$$rx + 18y = 20$$

In the given system of equations, r is a constant. The system has infinitely many solutions. What is the value of r ?

(A) 2

(B) 3

(C) 6

(D) 9

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666	0.66 .66 0.67 .67



14



Mark for Review

A club plans to sell tote bags. The club members estimate they will sell 80 tote bags when the bags are priced at \$9 each. For every price increase of \$1, they estimate they will sell 8 fewer bags. What is the estimated revenue, in dollars, when the bags are priced at \$12 each? (revenue = price $\times$ number sold)

Answer Preview:

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	3 1/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
	.6667	.66
	0.666	0.67
		.67



15



Mark for Review

$$3x + 6y = 17$$

$$-3x - 4y = 5$$

The solution to the given system of equations is (x, y) . What is the value of $2y$?

Answer Preview:

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .66 .6667 0.666	0.66 .66 0.67 .67



16

Mark for Review

What value of w is the solution to the equation $0.7w - 0.57 = 7(w - 0.001)$?

Answer Preview:



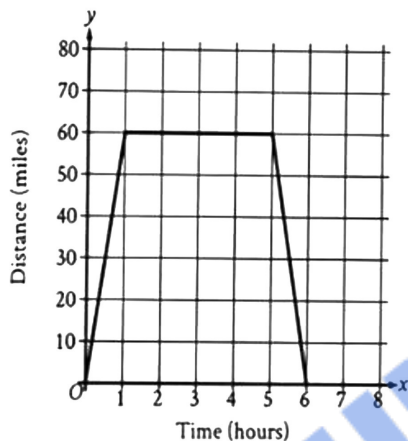
Which quadratic equation has exactly one distinct real solution?

(A) $x^2 - 16 = 0$

(B) $x^2 + 16 = 0$

(C) $x^2 - 16x + 56 = 0$

(D) $x^2 - 16x + 64 = 0$



Quinidra drove her car on a straight road to a certain location. The graph models Quinidra's distance, in miles, from her home as a function of time, in hours. Which of the following could the model describe?

- (A) Quinidra drove away from her home at a constant speed for 1 hour, spent 4 hours at the location, then continued to drive away from her home at a constant speed for 1 hour.
- (B) Quinidra drove away from her home at a constant speed for 1 hour, spent 4 hours at the location, then drove back to her home at a constant speed for 1 hour.
- (C) Quinidra drove away from her home at an increasing speed for 1 hour, spent 4 hours at the location, then drove back to her home at a decreasing speed for 1 hour.
- (D) Quinidra drove away from her home at an increasing speed for 1 hour, drove at a constant speed for 4 hours, then drove back to her home at a decreasing speed for 1 hour.

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .66 .6667 0.666 .67 0.667	0.66 .66 0.67 0.666 .67



19



Mark for Review

Mass(grams)	Frequency
10	12
20	6
30	8
40	9

The frequency table summarizes the mass, in grams, of 35 objects. What is the mean mass, in grams, of these objects?

Answer Preview:



What is the y -intercept of $y = -(4)^x - 39$ in the xy -plane?

(A) $(0, -40)$

(B) $(0, -43)$

(C) $(0, -39)$

(D) $(0, -38)$

Answer	Acceptable ways to enter answer	Unacceptable:will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666 0.667	0.66 .66 0.67 .67



21



Mark for Review

A rectangular poster has an area of 330 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 50%. What is the area of the copy, in square inches?

Answer Preview:



A circle in the xy -plane has center $(28, 5)$ and y -intercepts $(0, -16)$ and $(0, 26)$. What is the radius of this circle?

(A) 35

(B) 26

(C) 21

(D) 16

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Module-2

1



Mark for Review



The total cost $f(x)$, in dollars, to lease a car for 24 months from a particular car dealership is given by $f(x) = 24x + 1000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is \$300?

(A)

\$6,200

(B)

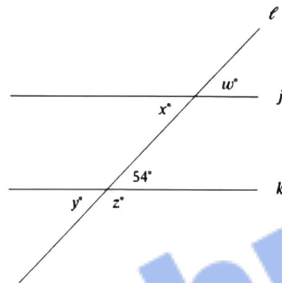
\$7,000

(C)

\$8,200

(D)

\$25,300



Note: Figure not drawn to scale.

In the figure shown, line l intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

(A) $w = 54$

(B) $x = 126$

(C) $y = 54$

(D) $z = 126$



$$5x + 3 = rx + 7$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

(A) 21

(B) 8

(C) 7

(D) 5



If $5(2x + 9) = 3(2x + 9) + 74$, what is the value of $2x + 9$?

(A) 74

(B) 37

(C) 28

(D) 14



If 350% of n is 35, what is the value of n ?

(A) 12,250

(B) 1,225

(C) 1,000

(D) 10

x	$h(x)$
4	17
6	23

The table shows two values of x and their corresponding values of $h(x)$ for the linear function h . Which equation defines h ?

(A) $h(x) = \frac{1}{3}x - \frac{5}{3}$

(B) $h(x) = 3x + 5$

(C) $h(x) = 4x + 17$

(D) $h(x) = 6x + 23$



$$y = x^2 + 17x + 4$$

$$y = x + 4$$

The graphs of the equations in the given system intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

(A) -16

(B) -15

(C) 13

(D) 15



A piece of string with a length of 108 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 8 more than 4 times the value of y . What is the value of x ?

(A) 25

(B) 29

(C) 58

(D) 88



A landscaper will plant two types of shrubs, hydrangeas and winter hazels, in a park. There will be no more than 244 total shrubs planted, and the number of hydrangeas planted will be at most three times the number of winter hazels planted. Which of the following systems of inequalities best represents this situation, where h is the number of hydrangeas that will be planted and w is the number of winter hazels that will be planted?

(A)
$$\begin{aligned} h + w &\leq 244 \\ h &\leq 3w \end{aligned}$$

(B)
$$\begin{aligned} h + w &\leq 244 \\ 3h &\geq w \end{aligned}$$

(C)
$$\begin{aligned} h + w &\geq 244 \\ h &\leq 3w \end{aligned}$$

(D)
$$\begin{aligned} h + w &\geq 244 \\ 3h &\geq w \end{aligned}$$



$$y \leq x + 3$$

$$y \geq -3x - 7$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

(A) $(0, -14)$

(B) $(0, 14)$

(C) $(-14, 0)$

(D) $(14, 0)$



$$3x^2 - 5x - 5 = 0$$

What is the greatest solution to the given equation?

(A) $\frac{5}{6} - \frac{\sqrt{85}}{6}$

(B) $\frac{5}{6} + \frac{\sqrt{85}}{6}$

(C) $\frac{5}{3} - \frac{\sqrt{85}}{3}$

(D) $\frac{5}{3} + \frac{\sqrt{85}}{3}$

Examples



Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	2/3	
	.6666	0.66
		.66
	.6667	0.67
	0.666	

12



Mark for Review

$$f(x) = -0.0038400x^2 + 15.236x - 15,103$$

The model shown gives the predicted average Arctic sea ice area $f(x)$, in millions of square kilometers, for September of each year x , where x represents a year and $1979 \leq x \leq 2012$. Based on the model, what is the positive difference, in millions of square kilometers, between the predicted average Arctic sea ice area for September of the year 1991 and the predicted average Arctic sea ice area for September of the year 1992? (Round your answer to the nearest thousandth.)

Answer Preview:



What is the y -intercept of the graph of $y = \left(\frac{8}{13}\right)^{x+1}$ in the xy -plane?

(A) $(8, 13)$

(B) $(0, 13)$

(C) $(0, 8)$

(D) $\left(0, \frac{8}{13}\right)$



$$f(x) = x^2 - 4x - 320$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

(A) $f(x) = x^2 - 4(x + 80)$

(B) $f(x) = (x - 2)^2 + (-324)$

(C) $f(x) = x(x - 4) + (-320)$

(D) $f(x) = (x + 16)(x - 20)$

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	31/2 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666	0.66 .66 0.67 --



15



Mark for Review

Line s in the xy -plane is defined by $6x + 2y = 0$. Line t is perpendicular to line s and passes through the points $(7, 0)$ and $\left(k, \frac{4}{3}\right)$. What is the value of k ?

Answer Preview:



$$g(x) = x^3 + ax^2 + bx + c$$

The function g is defined by the given equation, where a , b , and c are integer constants. The zeros of the function are -2 , -7 , and 6 . What is the value of a ?

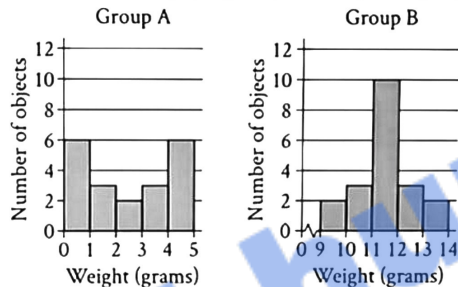
(A) -84

(B) -3

(C) 3

(D) 84

The weight of each object in two groups, A and B, was recorded. The histograms represent the distributions of weight, in grams, for the objects in groups A and B.



The mean weight of the objects in group A is equal to 2.1, and the mean weight of the objects in group B is equal to 11.1. Which of the following statements about the standard deviations of weight for the objects in these two groups is true?

- ☐ A The standard deviation of weight for the objects in group A is less than the standard deviation of weight for the objects in group B.
- ☐ B The standard deviation of weight for the objects in group A is equal to the standard deviation of weight for the objects in group B.
- ☐ C The standard deviation of weight for the objects in group A is greater than the standard deviation of weight for the objects in group B.
- ☐ D There is not enough information to compare the standard deviations of weight for the objects in these two groups.



A carpenter charges a flat rate of \$216 for the first 3 hours of work and \$60 for each additional hour of work. Which equation gives the total amount y , in dollars, that the carpenter charges for x hours of work, where $x > 3$?

(A) $y = 60x + 216$

(B) $y = 216x + 396$

(C) $y = 60x + 36$

(D) $y = 216x + 60$

$$\frac{46}{p} = \frac{46}{q} - \frac{46}{r} - \frac{46}{s}$$

The given equation relates the positive variables p , q , r , and s . Which of the following is equivalent to q ?

(A) $\frac{prs}{pr + ps + rs}$

(B) $\frac{prs}{46p + 46r + 46s}$

(C) $p + r + s$

(D) $46(p + r + s)$

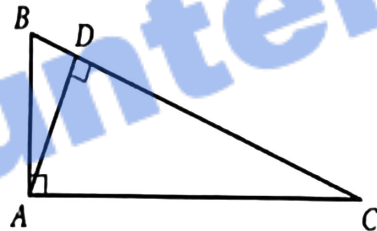
Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	$3\frac{1}{2}$
	$\frac{7}{2}$	$3\frac{1}{2}$
$\frac{2}{3}$	$\frac{2}{3}$	
	0.66	
	.6666	
	.66	
	.6667	
	0.67	
	0.666	
		.67



20



Mark for Review



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . The measure of $\angle ABD$ is w° , the measure of $\angle CAD$ is z° , and $AD = 178$. If $\cos w^\circ = \frac{9}{41}$, what is the value of $\tan z^\circ$?

Answer Preview:



A right square pyramid has a total surface area of 70,560 square inches, and the combined surface area of the four lateral faces of this pyramid is 38,160 square inches. What is the height, in inches, of this pyramid?

(A) 56

(B) 90

(C) 106

(D) 180

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5	
	3.50	31/2
	7/2	3 1/2
$\frac{2}{3}$	$\frac{2}{3}$	
	.6666	0.66
	.6667	.66

22



Mark for Review

$$\frac{1}{cx} = \frac{x}{96} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

Answer Preview: